

Matbench Discovery

A framework to evaluate machine learning crystal stability predictions

Janosh Riebesell^{1,2} Rhys E. A. Goodall¹ Philipp Benner³
 Yuan Chiang^{2,4} Bowen Deng^{2,4} Gerbrand Ceder^{2,4} Mark Asta^{2,4}
 Alpha A. Lee¹ Anubhav Jain² Kristin A. Persson^{2,4}

December 12, 2024

¹ Department of Physics, University of Cambridge, UK

² Lawrence Berkeley National Laboratory, USA

³ Federal Institute of Materials Research and Testing (BAM), Germany

⁴ Department of Materials Science and Engineering, University of California - Berkeley, USA

Abstract

The rapid adoption of machine learning (ML) in domain sciences necessitates best practices and standardized benchmarking for performance evaluation. We present Matbench Discovery, an evaluation framework for ML energy models, applied as pre-filters for high-throughput searches of stable inorganic crystals. This framework addresses the disconnect between thermodynamic stability and formation energy, as well as retrospective vs. prospective benchmarking in materials discovery. We release a Python package to support model submissions and maintain an online leaderboard, offering insights into performance trade-offs. To identify the best-performing ML methodologies for materials discovery, we benchmarked various approaches, including random forests, graph neural networks (GNNs), one-shot predictors, iterative Bayesian optimizers, and universal interatomic potentials (UIP). Our initial results rank models by test set F1 scores for thermodynamic stability prediction: EquiformerV2 + DeNS > Orb > SevenNet > MACE > CHGNet > M3GNet > ALIGNN > MEGNet > CGCNN > CGCNN+P > Wrenformer > BOWSR > Voronoi fingerprint random forest. UIPs emerge as the top performers, achieving F1 scores of 0.57–0.82 and discovery acceleration factors (DAF) of up to 6x on the first 10k stable predictions compared to random selection. We also identify a misalignment between regression metrics and task-relevant classification metrics. Accurate regressors can yield high false-positive rates near the decision boundary at 0 eV/atom above the convex hull. Our results demonstrate UIPs’ ability to optimize computational budget allocation for expanding materials databases. However, their limitations remain underexplored in traditional benchmarks. We advocate for task-based evaluation frameworks, as implemented here, to address these limitations and advance ML-guided materials discovery.

1 Introduction

The challenge of evaluating and benchmarking the rapid evolution of machine learning models is common to many domain sciences. Specifically, the lack of pre-agreed upon tasks and data sets can obscure the performance of the model, making comparisons difficult. Material science presents one such domain, where in the last decade, the number of machine learning publications and associated models have increased dramatically. Similar to other domains, such as drug discovery and protein design, the ultimate success is often associated with the discovery of a new material with specific functionality. In the combina-

torial sense, material science can be viewed as an optimization problem of mixing and arranging different atoms with a merit function that captures the complex range of properties that emerge. To date, $\sim 10^5$ combinations have been tested experimentally [1, 2], $\sim 10^7$ have been simulated [3–7], and upwards of $\sim 10^{10}$ possible quaternary materials are allowed by electronegativity and charge-balancing rules [8]. The space of quaternaries and higher is even less explored, leaving vast numbers of potentially useful materials to be discovered. The discovery of new materials is a key driver of technological progress and lies on the path to more efficient solar cells, lighter and longer-lived batteries, smaller and more efficient transistor gates just to name a few. In light of our sustainability goals, these advances cannot come fast enough. Any speed-up new discovery

⁰Correspondence to janosh.riebesell@gmail.com, kristinpersson@berkeley.edu

methods might yield should be leveraged to their fullest extent.

Despite significant advances in empirical, theoretical, and computational materials science, discovering new materials still requires complex calculations, labor-intensive trial-and-error experimentation, and often happens fortuitously rather than through rational design. Machine learning (ML) methods efficiently extract and distill trends from large datasets, and can handle high dimensionality, multiple objectives [9], uncertainty [10–12], and noisy or sparse data [13, 14], making them powerful additions to augment the traditional computational materials science tool set.

In particular, we focus on the role of ML to accelerate the use of Kohn-Sham density-functional theory (DFT) in the materials discovery pipeline. In comparison to other simulation frameworks DFT offers a compelling compromise between fidelity and cost that has seen it adopted as a workhorse method by the computational material science community. The great strengths of DFT as a methodology have led it to demand up to 45% of core hours at the UK-based Archer2 Tier 1 supercomputer [15] and over 70% allocation time in materials science sector at NERSC [16, 17]. This heavy resource requirement drives demand for ways to reduce or alleviate its computational burden, such as efficiency improvements or substitution from ML approaches.

ML models are less accurate and reliable but orders of magnitude faster than ab-initio simulation. This makes them most suitable for use in high-throughput (HT) searches to triage more expensive, higher-fidelity simulation methods such as DFT. The use of neural networks for learning the DFT potential energy surface (PES) can be traced as far back as [18]. This work kicked off rapid advances and significant efforts to fit ever more sophisticated ML models to known samples of the PES. Initially, most of these models were trained and deployed as interatomic potentials (also known as force fields) to study known materials of interest, a workflow that requires curating custom training data for each new system of interest [19, 20]. As larger and more diverse datasets have emerged from initiatives like the Materials Project (MP) [3], AFLOW [5] or the Open Quantum Materials Database (OQMD) [4], researchers have begun to train so-called universal models that cover 90 or more of the most-application relevant elements in the periodic table. This opens up the prospect of ML-guided materials discovery to increase the hit rate of stable crystals and speed up DFT- and expert-driven searches.

Progress in ML for materials is often measured according to performance on standard benchmark datasets. As ML models have grown in complexity and applicability, benchmark datasets need to grow with them to accurately measure their usefulness. However, due to the rapid pace of the field and the variety of possible approaches for framing the discovery problem, no large-scale benchmark yet exists for measuring the ability of ML to accelerate mate-

rials discovery. As a result, it is unclear which methodologies or models are best suited for this task. Recent works focusing on prospective computational materials discovery have proposed strategies based on one-shot coordinate free predictors [11], iterative Bayesian optimizers [21], and universal interatomic potentials (UIP) [22–24]. These papers deploy their respective models on specific systems and custom datasets to highlight certain strengths but have yet to be compared in a systematic standardized manner that allows them to be ranked by their ability to accelerate materials discovery. Our work aims to identify the state-of-the-art (SOTA) model by proposing a novel evaluation framework that closely simulates a real-world discovery campaign guided by ML models. Our findings show that UIPs outperform all other methodologies we tested both in accuracy and robustness.

We hope that creation of benchmarks following this framework create a pathway through which interdisciplinary researchers with limited material science backgrounds can contribute usefully to model architecture and methodology development on a relevant task and thereby aid progress in Material Science.

2 Evaluation Framework for Materials Discovery

In this work we proposed a benchmark task that addresses four central challenges that we believe to be requirements when seeking to justify experimental validation of ML predictions:

1. **Prospective Benchmarking:** Idealized and overly simplified benchmarks can fail to reflect the real-world challenges a model faces when used in an actual discovery campaign. This can result in a disconnect between benchmark metrics and real-world performance. Possible reasons for this include choosing the wrong target [14] or picking an unrepresentative training and test splits [25, 26]. For small datasets of materials properties, ‘Leave-Out’ data splitting strategies are often used to assess model performance [27–29]. However, in our target domain large quantities of diverse data ($\sim 10^5$) are available and hence retrospective splitting strategies predicated on clustering can end up testing artificial or unrepresentative use cases. This encourages using new sources of prospectively generated test data to understand application performance. Adopting this principle, the intended discovery workflow should be used to generate the test data leading to a substantial but realistic covariate shift between the training and test distributions that gives a much better indicator likely of performance on additional application of the same discovery workflow.

Relevant Targets: In the case of materials discovery, high-throughput DFT calculated formation en-

ergies - although widely used as regression targets - do not indicate thermodynamic stability or synthesizability. In contrast, the thermodynamic competition between one material and its competing phases in the same chemical system determines its thermodynamic stability, which can be quantified by the material’s energy to the convex hull phase diagram. The distance to the convex hull is the leading order term and strongly predictive of (meta-)stability at standard conditions [30]. Whilst other factors such as finite temperature effects, vibrational, and configurational entropy can modify a material’s stability under real-world conditions they are prohibitive to calculate in high-throughput workflows, leaving the distance to the convex hull as the best signal-vs-data-availability compromise to use as a target.

Moreover, ML models relying on relaxed crystal structures as input render any discovery pipeline circular since obtaining relaxed structures requires computationally expensive DFT simulations, thereby depending on the very process we intend to accelerate.

2. **Informative Metrics:** Global error metrics like MAE, RMSE and R^2 can often give practitioners a false sense of security about the model’s reliability. For example, accurate regressors are susceptible to unexpectedly high false-positive rates if those nominally accurate predictions lie close to the decision boundary and the failed predictions can incur a high opportunity cost by wasting lab time and resources. Consequently, models should be measured on their ability to lead to correct decision making patterns not regression accuracy. One way to do this is by defining a selection criteria and then evaluating regression models primarily in terms of their classification performance.
3. **Scalability:** Future materials discovery efforts are likely to target broad chemical spaces and large data regimes. Small benchmarks can lack chemical diversity, and obfuscate poor scaling relations or weak out-of-distribution (OOD) performance. For instance, random forests achieve excellent performance on small datasets but are typically outperformed by neural networks on large datasets due to the benefits of representation learning [31]. Whilst we propose that large training sets are necessary to adequately differentiate the ability of models to learn in the larger data regime, given the enormous size of the configurational space of materials yet to be explored we also propose that it is important to construct a task where the test set is larger than the training set to mimic true deployment at scale. No other inorganic materials benchmarks test the prospects of large-scale deployment in this manner.

We highlight two specific benchmarking efforts that have partially addressed the above challenges: Matbench

[32] and the Open Catalyst Project (OCP) [33]. Other valuable efforts such as MatSciML [34] and JARVIS-Leaderboard [35] aggregate a wide variety of material science related benchmark tasks, including from Matbench and OCP, but do not introduce distinct benchmarking design patterns to those seen in Matbench or the OCP.

By providing a standardized collection of 13 datasets ranging in size from ~ 300 to $\sim 132,000$ samples from both DFT and experimental sources, Matbench addresses the scalability challenge, highlighting how model performance changes as a function of data regime. Matbench helped focus the field of ML for materials, increase comparability across papers and provide a quantitative measure of progress in the field. Importantly, all tasks were exclusively concerned with the properties of known materials. We believe a task that simulates a materials discovery campaign by requiring materials stability prediction from unrelaxed structures to be a missing piece here.

OCP is a large-scale initiative aimed at discovering substrate-adsorbate combinations that can catalyze critical industrial reactions, transforming these adsorbates into more useful products. The OCP has released two datasets thus far, OCP20 [33] and OCP22 [36] for training and benchmarking ML models. OCP certainly addressed challenge 1 of closely mimicking a real-world problem. They have recently shown that despite not reaching their target accuracy to entirely replace DFT, using ML in conjunction with confirmatory DFT calculations dramatically speeds up their combinatorial screening workflow [37]. The team behind the OCP has a second initiative targeting materials for direct air capture called OpenDAC that has shared the ODAC23 dataset [38]. The OpenDAC benchmark is setup identically to the OCP.

We believe that addressing these four challenges will result in benchmarks that enable future ML-guided discovery efforts to confidently select appropriate models and methodologies for the expansion of computational materials databases.

2.1 Matbench Discovery

We propose a novel evaluation framework that places no constraints on the type of data a model is trained on as long as it would be available to a practitioner conducting a real materials discovery campaign. This means that for the high-throughput DFT data considered, any subset of the energies, forces, stresses or any other properties that can be routinely extracted from DFT calculations, such as magnetic moments, are valid training targets. All of these would be available to a practitioner performing a real materials discovery campaign and hence are permitted for training any model submission. We only enforce that at test time, all models must make predictions on the convex hull distance of the relaxed structure with only the unrelaxed structure as input. This setup avoids circularity in the discovery process, as unrelaxed structures can be cheaply enumerated through elemental substitution

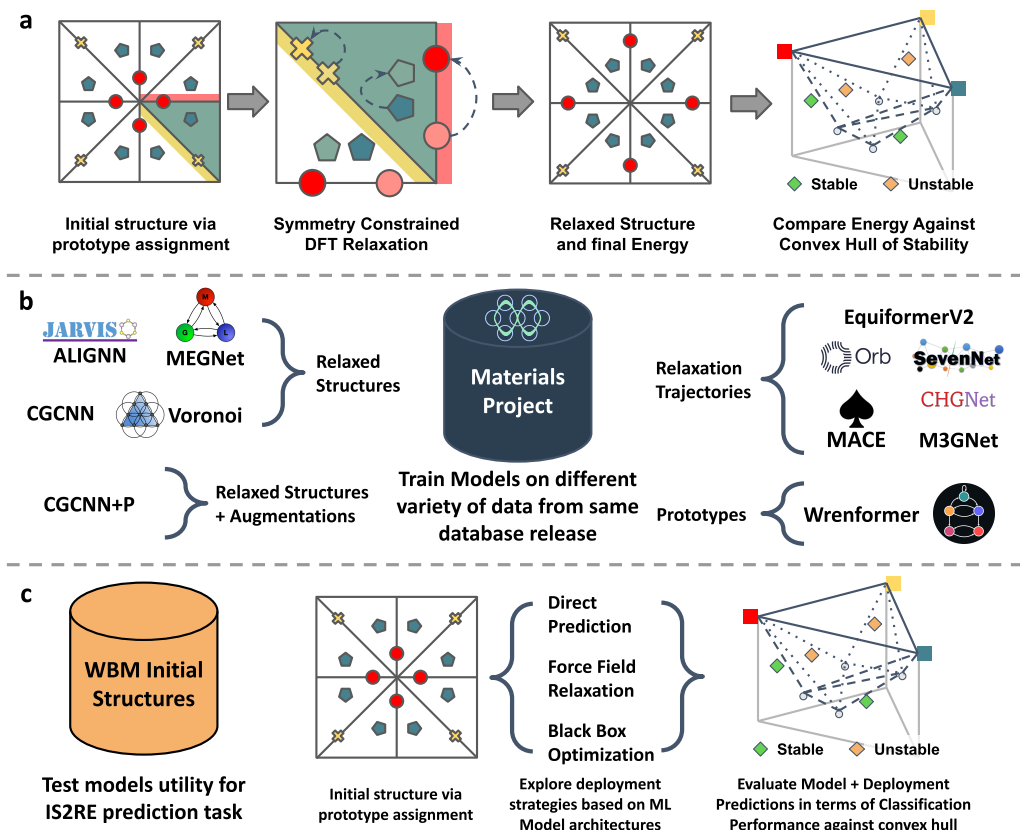


Figure 1: An overview of how data is used in Matbench-Discovery. a) shows a conventional prototype-based discovery workflow where different elemental assignments to the sites in a known prototype are used to create a candidate structure. This candidate is relaxed using DFT to arrive at a relaxed structure that can be compared against a reference convex hull. This sort of workflow was used to construct the WBM data set. b) highlights that databases such as the Materials Project provide a rich set of data which different academic groups have used to explore different types of models. While prior work tended to focus on individual modalities, our framework enables consistent model comparisons across modalities. c) shows the proposed test evaluation framework where the end user takes a machine learning model and uses it to predict a relaxed energy given an initial structure (IS2RE). This energy is then used to make a prediction as to whether the material will be stable or unstable with respect to a reference convex hull. From an applications perspective, this classification performance is better aligned with intended use cases in screening workflows.

methodologies and do not contain information inaccessible in a prospective discovery campaign. Figure 1 provides a visual overview of design choices.

The convex hull distance of a relaxed structure is chosen as the measure of its thermodynamic stability, rather than the formation energy, as it informs the decision on whether to pursue a potential candidate crystal. This decision was also motivated by [14] who found that even composition-only models are capable of predicting DFT formation energies with useful accuracy. However, when tasking those same models with predicting decomposition enthalpy, performance deteriorated sharply. This insight exposes how ML models are much less useful than DFT for discovering new inorganic solids than would be expected given their low prediction errors for formation energies due to the impact of random as opposed to systematic errors.

Standard practice in ML benchmarks is to hold all vari-

ables fixed – most importantly the training data – and vary only the model architecture to isolate architectural effects on the performance. We deliberately deviate from this practice due to diverging objectives from common ML benchmarks. Our goal is to identify the best methodology for accelerating materials discovery. What kind of training data a model can ingest is part of its methodology. Unlike energy-only models, UIPs benefit from the additional training data provided by the forces and stresses recorded in DFT relaxations. This allows them to learn a fundamentally higher fidelity model of the physical interactions between ions. That is a genuine advantage of the architecture and something any benchmark aiming to identify the optimal methodology for materials discovery must reflect. In light of this utilitarian perspective, our benchmark contains models trained on varying datasets, and any model that can intake more physical modalities

from DFT calculations is a valid model for materials discovery.

We define the Materials Project (MP) [3] v2022.10.28 database release as the maximum allowed training set for any compliant model submission. Models may train on the complete set of relaxation frames, or any subset thereof such as the final relaxed structures. Any subset of the energies, forces, and stresses are valid training targets. In addition any auxiliary tasks such as predicting electron densities, magnetic moments, site-partitioned charges, etc. that can be extracted from the output of the DFT calculations are allowed for multi-task learning [39]. Our test set consists of the unrelaxed structures in the WBM dataset [40]. Their target values are the PBE formation energies of the corresponding DFT-relaxed structures.

2.1.1 Materials Project Training Set

The Materials Project is a widely-used database of inorganic materials properties that have been calculated using high-throughput ab-initio methods. At the time of writing, the Materials Project database [3] has grown to ~ 154 k crystals, covering diverse chemistries and providing relaxed and initial structures as well as the relaxation trajectory for every entry.

Our benchmark defines the training set as all data available from the v2022.10.28 MP release. We recorded a snapshot of energies, forces, stresses and magnetic moments for all MP ionic steps on 2023-03-15 as the canonical training set for Matbench Discovery, and provide convenience functions through our Python package for easily feeding that data into future model submissions to our benchmark.

Flexibility in specifying the dataset allows authors to experiment with and fully exploit the available data. This choice is motivated by two factors. First, it may seem that models trained on multiple snapshots containing energies, forces, and stresses receive more training data than models trained only on the energies of relaxed structures. However, the critical factor is that all this additional data was generated as a byproduct of the workflow to produce relaxed structures. Consequently, all models are being trained using data acquired at the same overall cost. If some architectures or approaches can leverage more of this byproduct data to make improved predictions this is a fair comparison between the two models. This approach diverges philosophically from other benchmarks such as the OCP and Matbench where it has been more common to subcategorize different models and look at multiple alternative tasks (e.g. composition-only vs structure-available in Matbench or IS2RS, IS2RE, S2EF in OCP) and do not make direct comparisons of this manner. Second, recent work in the space from [41, 42] has claimed that much of the data in large databases like MP are redundant and that models can be trained more effectively by taking a subset of these large data pools. From a systems-level perspective, identifying novel cleaning or active-learning strate-

gies to make better use of available data may be as crucial as architectural improvements, as both can similarly enhance performance, especially given the prevalence of errors in high-throughput DFT. Consequently, such strategies where they lead to improved performance should be able to be recognized within the benchmark. We encourage submissions to submit ablation studies showing how different system-level choices affect performance. Another example of a system-level choice that may impact performance is the choice of optimizer, for example FIRE [43] vs L-BFGS, in the relaxation when using UIP models.

We highlight several example datasets that are valid within the rules of the benchmark that take advantage of these freedoms. The first is the MP-crystals-2019.4.1 dataset [44] which is a subset of 133,420 crystals and their formation energies that form a subset of the v2021.02.08 MP release. The MP-crystals-2022.10.28 dataset is introduced with this work comprising a set of 154,719 structures and their formation energies drawn from the v2021.02.08 MP release. The next is the MPF.2021.2.8 dataset [22] curated to train the M3GNet model which takes a subset of 62,783 materials from the v2021.02.08 MP release. The curators of the MPF.2021.2.8 dataset down-sampled the v2021.02.08 release significantly to select a subset of calculations that they believed to be most self-consistent. Rather than taking every ionic step from the relaxation trajectory this dataset opts to select only the initial, final and one intermediate structure for each material to avoid biasing the dataset towards examples where more ionic steps were needed to relax the structure. Consequently the dataset consists of 188,349 structures. The MPF.2021.2.8 is a proper subset of the training data as no materials were deprecated between the v2021.02.08 and v2022.10.28 database releases. The final dataset we highlight, with which several of the UIP models have been trained, is the MPtrj dataset [23]. This dataset was curated from the earlier v2021.11.10 MP release. The MPtrj dataset is a proper subset of the allowed training data but several potentially anomalous examples from within MP were cleaned out of the dataset before the frames were sub-sampled to remove redundant frames. It is worth noting that the v2022.10.28 release contains a small number of additional Perovskite structures not found in MPtrj that could be added to the training set within the scope of the benchmark.

We note that the v2023.11.1 deprecated a large number of calculations so data queried from subsequent database releases is not considered valid for this benchmark.

2.1.2 WBM Test Set

The WBM dataset [40] consists of 257,487 structures generated via chemical similarity-based elemental substitution of MP source structures followed by DFT relaxation and calculating each crystal’s convex hull distance. The element substitutions applied to a given source structure were determined by random sampling according to the